



Chaehyun Lim

Atlanta, GA | +1 404 642 4638 | clim70@gatech.edu | F-1 Visa (CPT / OPT Eligible)

[linkedin.com/in/lim-chaehyun](https://www.linkedin.com/in/lim-chaehyun)

Hardware Integration & Test

Aerospace B.S./M.S. engineer with hands-on circuit design, electronics instrumentation, hardware integration, and bench/flight validation across UAV and electric-aircraft projects.

EDUCATION

Georgia Institute of Technology, Atlanta, GA

- M.S. Aerospace Engineering Expected May 2027
- B.S. Aerospace Engineering | GPA 3.65 / 4.00 | Graduated with Highest Honor May 2026
- **Relevant Coursework:** Circuits & Electronics · Instrumentation & Electronics Lab

HARDWARE INTEGRATION & TEST EXPERIENCE

Moore Dynamics & Analytics Lab, Georgia Tech

Oct 2024 – May 2026

Undergraduate Researcher — Automated Test Systems

- Integrated a **UR30 robot arm**, Arduino-controlled stepper motor, and **NI cDAQ** chain into an automated modal-impact test system, replacing manual impact strikes.
- Built a **MATLAB data-acquisition GUI** for test execution and logging; supported repeatable controlled impact loading from **250 N to 3,000 N**.

CubeSat Reaction-Wheel Heading Control, Georgia Tech

Jan 2026 – May 2026

Senior Capstone — Prototype Build & Validation

- **Ranked #1 in class; selected as a Top 5 capstone project.** Created the system circuit schematic in **KiCad** and integrated Arduino Nano, MPU-6050 IMU, ESC, BLDC motor, and LiPo power into a reaction-wheel control prototype.
- Built a **MATLAB/Simulink 1-DOF reaction-wheel model** and executed hardware validation against quantitative targets (**2.7 s rise, 5.9 s settling, <8.24% overshoot, $\pm 2.8^\circ$ heading error**).

43rd VFS Student Design Competition — Hybrid-Electric Tiltrotor

Aug 2025 – Jun 2026

Propulsion Lead

- Developed the hybrid-electric power architecture and built **MATLAB mission-energy models** to set the turbogenerator/battery power split across hover, conversion, and cruise.
- Team placed **2nd globally in the undergraduate division**; coordinated propulsion interfaces and deliverables with aerodynamics and structures.

Forest Fire Detection UAV, Georgia Tech

Aug 2024 – Dec 2025

Pilot & System Designer

- Integrated UAV structure, a 3D-printed battery enclosure, and thermal/optical sensor payloads; executed waypoint flight tests to verify airframe stability and payload function.

Vertical Lift Research Center of Excellence, Georgia Tech

Jan 2025 – May 2025

Undergraduate Researcher — Battery Testing & Degradation Analysis

- Ran Li-polymer charge/discharge cycle-life tests across C-rate and temperature conditions; analyzed capacity fade and internal-resistance growth for AAM/eVTOL applications.

LEADERSHIP

ROK Defense Intelligence Command — Open-Source Center (military service)

Oct 2020 – Apr 2022

Senior Squad Leader / Intelligence Editor in Chief

- Led an **11-person squad** and authored a **100-page operational guideline**; selected as a 2021 U.S.–ROK CCPT command interpreter and **recognized by the Chairman of the ROK JCS**.

SKILLS & CERTIFICATIONS

Electrical & Test: Oscilloscope · Digital multimeter · Bench power supply · Function generator · Bench testing · Circuit debugging · Verification & validation · Sensor/DAQ integration

Hardware & Tools: KiCad (schematic capture) · Arduino · NI cDAQ · MATLAB/Simulink · SolidWorks · 3D printing · PWM/ESC/BLDC system exposure · Python (coursework)

Certifications: FAA Part 107 · EASA A1/A3 · ROK Unmanned Multicopter Pilot Certificate (20+ logged flight hours)

Languages: Korean (native) · English (fluent) · Chinese (HSK 5)